

# CPSC 304 Project Cover Page

Milestone # : 4

Date : 28-November-2025

Group Number : 73

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

## MILESTONE 4

### **Project Description**

This project aims to act as a centralized database for a theme park designed to integrate ticketing, hotel reservation systems, loyalty programs, and ride operations. By consolidating all customer and operational data into one platform, it enables staff and management to efficiently query customer data.

This project included creating an ER diagram to model the data, creating a relational schema based on this ER diagram, performing data normalization, and writing SQL DDL (data description language) for the normalized schema.

The final stage of our project involved creating a web application which allows users to view Customer data, and perform various queries on this data, such as finding which customers have rode all rides at the theme park, finding the average number of points by gender, finding how many total guests have visited the theme park in each calendar month and more. The user can also perform basic operations, such as inserting new Customers into the database, updating customer information, removing customers, filtering customers based on their attributes, and viewing only a subset of Customer attributes such as name, birthdate, etc.

### **Differences Between the Final Schema and Milestone 2 Schema**

The final schema is nearly the exact same as the one we turned in, aside from changing the “Sex” attribute in Customer to the more broad attribute of “Gender”. This was done in order to allow more groups, enabling our Query 7 (Aggregation with GROUP BY) to have sufficiently complex output.

### **SQL Queries along with File names and line numbers where the given query can be found**

#### ***Query 1 : INSERT Values into Customer***

*File Name: q1\_insert\_customer.sql (found in team\_73 / sql\_commands / q1)*

*Line Number: 1*

```
INSERT INTO Customer (CustomerID, CustomerName, DOB, Sex)
VALUES (:customerID, :customerName, TO_DATE(:dateOfBirth, 'YYYY-MM-DD'), :sex)
```

*File Name: q1\_insert\_guest.sql (found in team\_73 / sql\_commands / q1)*

*Line Number: 1*

```
INSERT INTO Guest (CustomerID, DateOfVisit)
VALUES (:customerID, TO_DATE(:dateOfVisit, 'YYYY-MM-DD'))
```

*File Name: q1\_insert\_loyalty\_member.sql (found in team\_73 / sql\_commands / q1)*

*Line Number: 1*

```
INSERT INTO LoyaltyMember (CustomerID, LoyaltyID, Points, UUID)
VALUES (:customerID, :loyaltyID, :loyaltyPoints, :uuid)
```

### **Query 2: UPDATE Values in Loyalty Member**

*File Name: q2\_update\_customer\_base.sql (found in team\_73 / sql\_commands / q2)*

*Line Number: 1*

```
UPDATE Customer SET {{set_clauses}} WHERE CustomerID = :customerID
```

*File Name: q2\_update\_seasonpass\_loyalty.sql (found in team\_73 / sql\_commands / q2)*

*Line Number: 1*

```
UPDATE SeasonPass2
SET LoyaltyID = :newLoyaltyID
WHERE LoyaltyID = :oldLoyaltyID
```

*File Name: q2\_merge\_loyalty\_member\_partial.sql (found in team\_73 / sql\_commands / q2)*

*Line Number: 1*

```
UPDATE LoyaltyMember SET
    LoyaltyID = NVL(:loyaltyID, LoyaltyID),
    Points = NVL(:loyaltyPoints, Points)
WHERE
    CustomerID = :CustomerID
```

### **Query 3: DELETE Values in Loyalty Member**

*File Name: q3\_delete\_loyalty\_member.sql (found in team\_73 / sql\_commands / q3)*

*Line Number: 1*

```
DELETE FROM Customer WHERE CustomerID = :customerID
```

### **Query 4: Selection in Customer**

*File Name: q4\_customer\_selection\_base.sql (found in team\_73 / sql\_commands / q4)*

*Line Number: 1*

```
SELECT
    c.CustomerID,
    c.CustomerName,
    c.Sex,
    TO_CHAR(c.DOB, 'YYYY-MM-DD') AS DOB,
    TO_CHAR(g.DateOfVisit, 'YYYY-MM-DD') AS DateOfVisit,
    lm.LoyaltyID,
    lm.Points
FROM Customer c
```

```
LEFT JOIN Guest g ON g.CustomerID = c.CustomerID
LEFT JOIN LoyaltyMember lm ON lm.CustomerID = c.CustomerID
```

#### **Query 5: Projection on Customer**

*File Name: [q5\\_select\\_customer\\_profiles.sql](#) (found in team\_73 / sql\_commands / q5)*

*Line Number: 1*

```
SELECT
  c.CustomerID,
  c.CustomerName,
  c.Sex,
  TO_CHAR(c.DOB, 'YYYY-MM-DD') AS DOB,
  TO_CHAR(g.DateOfVisit, 'YYYY-MM-DD') AS DateOfVisit,
  lm.LoyaltyID,
  lm.Points
FROM Customer c
LEFT JOIN Guest g ON g.CustomerID = c.CustomerID
LEFT JOIN LoyaltyMember lm ON lm.CustomerID = c.CustomerID
ORDER BY c.CustomerID
```

#### **Query 6: Join**

*File Name: [q6\\_customers\\_by\\_ride.sql](#) (found in team\_73 / sql\_commands / q6)*

*Line Number: 1*

```
SELECT DISTINCT
  c.CustomerID,
  c.CustomerName,
  CASE
    WHEN lm.CustomerID IS NOT NULL THEN 'Loyalty Member'
    ELSE 'Guest'
  END AS MembershipStatus
FROM ForRide fr
JOIN Ticket2 t ON t.TicketID = fr.TicketID
JOIN Booking2 b ON b.BookingNumber = t.BookingNumber
JOIN Customer c ON c.CustomerID = b.CustomerID
LEFT JOIN LoyaltyMember lm ON lm.CustomerID = c.CustomerID
WHERE fr.RideName = :rideName
AND b.CustomerID IS NOT NULL
ORDER BY c.CustomerName
```

#### **Query 7: Aggregation with GROUP BY: Average Number of Points Grouped by Gender/Sex**

*File Name: [q7\\_avg\\_points\\_group\\_genders\\_.sql](#) (found in team\_73 / sql\_commands / q7)*

*Line Number: 1*

**Description:** Finds the average number of points held by loyalty members of each gender, where the possible genders in the system are: Male, Female, Non-binary, Genderqueer.

```
SELECT
  C.Sex AS gender,
  ROUND(AVG(L.Points), 2) AS average_points
FROM Customer C
JOIN LoyaltyMember L ON L.CustomerID = C.CustomerID
GROUP BY C.Sex
ORDER BY C.Sex
```

**Query 8: Aggregation with HAVING:**

*File Name: q8\_guest\_counts\_by\_month.sql (found in team\_73 / sql\_commands / q8)*

*Line Number: 1*

**Description:** Find the total number of Guests that have visited in each calendar month (combined across all years), only including months that have had at least two cumulative Guests.

```
SELECT
  EXTRACT(MONTH FROM g.DateOfVisit) AS visit_month,
  COUNT(*) AS guest_count
FROM Guest g
GROUP BY EXTRACT(MONTH FROM g.DateOfVisit)
HAVING COUNT(*) > 1
ORDER BY visit_month
```

**Query 9: Nested Aggregation with GROUP BY: Year of Birth with the Minimum Average number of Points grouped by Year of Birth (Which birth year has the lowest average number of points)**

*File Name: q9\_min\_avg\_points\_by\_birth\_year.sql (found in team\_73 / sql\_commands / q9)*

*Line Number: 1*

**Description:** This query finds the year or years of birth for which the average number of points held by loyalty members born in that year is the lowest among all birth years. The query only considers the years that have more than one loyalty member born during that year.

```
WITH avg_points_per_year AS (
  SELECT
    EXTRACT(YEAR FROM c.DOB) AS birth_year,
    AVG(lm.Points) AS avg_points
  FROM Customer c
  JOIN LoyaltyMember lm ON lm.CustomerID = c.CustomerID
  WHERE c.DOB IS NOT NULL
  GROUP BY EXTRACT(YEAR FROM c.DOB)
  HAVING COUNT(*) > 1
```

```
)  
SELECT birth_year, avg_points  
FROM avg_points_per_year  
WHERE avg_points = (SELECT MIN(avg_points) FROM avg_points_per_year)  
ORDER BY birth_year
```

#### **Query 10: Division**

*File Name: q10\_customers\_who\_ride\_all\_rides.sql (found in team\_73 / sql\_commands / q10)*

*Line Number: 1*

**Description:** Finds the Customers that have Rode all Rides

```
WITH ride_counts AS (  
    SELECT COUNT(*) AS ride_total FROM Ride  
)  
customer_rides AS (  
    SELECT DISTINCT b.CustomerID, fr.RideName  
    FROM Booking2 b  
    JOIN Ticket2 t ON t.BookingNumber = b.BookingNumber  
    JOIN ForRide fr ON fr.TicketID = t.TicketID  
)  
SELECT c.CustomerID, c.CustomerName  
FROM Customer c  
JOIN customer_rides cr ON cr.CustomerID = c.CustomerID  
WHERE (SELECT ride_total FROM ride_counts) > 0  
GROUP BY c.CustomerID, c.CustomerName  
HAVING COUNT(DISTINCT cr.RideName) = (SELECT ride_total FROM ride_counts)  
ORDER BY c.CustomerID
```

#### **USE OF AI TOOLS IN MILESTONE 4**

No AI tools were used to assist in creating our submission for Milestone 4

# Project Description (Schema & Initial Data)

We have attached the relational schema and screenshots of the data tables of each relation after the initial SQL script is run.

## **RELATIONAL SCHEMA**

MaintenanceRecord1(MaintenancePerformed: VARCHAR, NumberOfWorkers: INTEGER)

- Primary Key: MaintenancePerformed

MaintenanceRecord2(RideName: VARCHAR, RecordID: INTEGER, MaintenancePerformed: VARCHAR, MaintenanceDate: DATE)

- Primary Key: RideName, RecordID
- Foreign Key: RideName

*\*This colour & underline means partial key as google sheets does not allow dotted underlines.*

Ride(RideName: VARCHAR, Capacity: INTEGER, InstallDate: DATE)

- Primary Key: RideName

Ticket1(ValidFrom: DATE, ValidUntil: DATE, ValidHours: INTEGER)

- Primary Key: ValidFrom, ValidHours

Ticket2(TicketID: INTEGER, **BookingNumber**: INTEGER, **ValidFrom**: DATE, RemainingUses: INTEGER, **ValidHours**: INTEGER)

- Primary Key: TicketID
- Foreign Key: BookingNumber, ValidFrom, ValidHours

FastPassTicket(**TicketID**: INTEGER, NumberOfRides: INTEGER, ValidHours: INTEGER)

TicketID is foreign key

- Primary Key: TicketID
- Foreign Key: TicketID

Customer(CustomerID: INTEGER, CustomerName: VARCHAR, Sex: VARCHAR, DOB: DATE)

- Primary Key: CustomerID

Guest(**CustomerID**: INTEGER, DateOfVisit: DATE)

- Primary Key: CustomerID
- Foreign Key: CustomerID

LoyaltyMember(**CustomerID**: INTEGER, LoyaltyID: INTEGER, Points: INTEGER)

- Primary Key: CustomerID
- Candidate Key: LoyaltyID
- Foreign Key: CustomerID

Booking1(BookingDate: DATE, Cost: INTEGER)

- Primary Key: BookingDate

Booking2(BookingNumber: INTEGER, BookingDate: DATE, **CustomerID**: INTEGER)

- Primary Key: BookingNumber

- Foreign Key: CustomerID (NOT NULL)

HotelStay(HotelBookingID: INTEGER, NumGuests: INTEGER, CheckInDate: DATE, CheckOutDate: DATE, **HotelName**: VARCHAR, **RoomName**: VARCHAR, **BookingNumber**: INTEGER)

- Merged with BookedAt, AtRoomType, HasHotelStay
- Primary Key: HotelBookingID
- Foreign Key: HotelName (NOT NULL), RoomName (NOT NULL), BookingNumber (NOT NULL)

RoomType1(BaseRate: FLOAT, Category: VARCHAR)

- Primary Key: Category

RoomType2(RoomName: VARCHAR, MaxGuests: INTEGER, Category: VARCHAR)

- Primary Key: RoomName

Hotel1(City: VARCHAR, Province: VARCHAR, PostalCode: VARCHAR)

- Primary Key: PostalCode

Hotel2(HotelName: VARCHAR, StreetAddress: VARCHAR, PostalCode: VARCHAR)

- Primary Key: HotelName

RateModifier(RateCode: INTEGER, Modifier: FLOAT)

- Primary key: RateCode

SeasonPass1(SeasonPassLevel: VARCHAR, SeasonStart: DATE, SeasonEnd: DATE,)

- Primary Key: SeasonPassLevel, SeasonStart

SeasonPass2(UUID: INTEGER, SeasonPassLevel: VARCHAR, SeasonStart: DATE, **LoyaltyID**: INTEGER)

- Primary Key: UUID
- Foreign Key: LoyaltyID (NOT NULL, UNIQUE)

ForRide(**RideName**: VARCHAR,

**TicketID**: INTEGER)

- Primary Key: RideName, TicketID
- Foreign Key: RideName, TicketID

OffersRoomType(**HotelName**: VARCHAR,

**RoomName**: VARCHAR,

Quantity: INTEGER)

- Primary Key: HotelName, RoomName
- Foreign Key: HotelName, RoomName

BookedWithRateModifier(**RateCode**: INTEGER,

**BookingNumber**: INTEGER)

- Primary Key: RateCode, BookingNumber
- Foreign Key: RateCode, BookingNumber

SeasonPassSpecial(**UUID**: INTEGER,

**RateCode**: INTEGER)

- Primary Key: UUID, RateCode
- Foreign Key: UUID, RateCode



**BOOKEDWITHRATEMODIFIER**

| RATECODE | BOOKINGNUMBER |
|----------|---------------|
| 0        | 1005          |
| 1        | 1001          |
| 2        | 1002          |
| 3        | 1003          |
| 4        | 1004          |

**BOOKING1**

| BOOKINGDATE         | COST |
|---------------------|------|
| 2025-07-10 00:00:00 | 120  |
| 2025-07-11 00:00:00 | 95   |
| 2025-07-12 00:00:00 | 90   |
| 2025-07-14 00:00:00 | 150  |
| 2025-07-16 00:00:00 | 110  |
| 2025-07-18 00:00:00 | 200  |
| 2025-08-01 00:00:00 | 130  |
| 2025-09-01 00:00:00 | 135  |
| 2025-09-14 00:00:00 | 140  |
| 2025-09-20 00:00:00 | 160  |

**BOOKING2**

| BOOKINGNUMBER | BOOKINGDATE         | CUSTOMERID |
|---------------|---------------------|------------|
| 1001          | 2025-07-10 00:00:00 | 1          |
| 1002          | 2025-08-01 00:00:00 | 2          |
| 1003          | 2025-09-01 00:00:00 | 3          |
| 1004          | 2025-07-16 00:00:00 | 4          |
| 1005          | 2025-07-18 00:00:00 | 5          |
| 1111          | 2025-07-10 00:00:00 | 1          |
| 1112          | 2025-07-10 00:00:00 | 1          |
| 1113          | 2025-07-11 00:00:00 | 1          |
| 1114          | 2025-07-12 00:00:00 | 1          |
| 1115          | 2025-07-12 00:00:00 | 1          |
| 2223          | 2025-09-14 00:00:00 | 3          |
| 2224          | 2025-09-14 00:00:00 | 3          |
| 2225          | 2025-09-14 00:00:00 | 3          |
| 2226          | 2025-09-14 00:00:00 | 3          |
| 2227          | 2025-09-20 00:00:00 | 3          |
| 2323          | 2025-07-18 00:00:00 | 5          |
| 2424          | 2025-07-18 00:00:00 | 5          |
| 2525          | 2025-07-18 00:00:00 | 5          |
| 2626          | 2025-07-18 00:00:00 | 5          |
| 2727          | 2025-07-18 00:00:00 | 5          |
| 7433          | 2025-09-01 00:00:00 | 8          |
| 7434          | 2025-09-01 00:00:00 | 8          |
| 7435          | 2025-09-01 00:00:00 | 8          |
| 7436          | 2025-09-01 00:00:00 | 9          |
| 7437          | 2025-09-01 00:00:00 | 9          |
| 7438          | 2025-09-01 00:00:00 | 10         |

**CUSTOMER**

| CUSTOMERID | CUSTOMERNAME       | SEX         | DOB                 |
|------------|--------------------|-------------|---------------------|
| 1          | Alice Johnson      | F           | 1990-05-14 00:00:00 |
| 2          | Bob Smith          | M           | 1988-08-09 00:00:00 |
| 3          | Carol White        | F           | 1995-11-02 00:00:00 |
| 4          | David Brown        | M           | 2000-01-25 00:00:00 |
| 5          | Eve Davis          | F           | 1993-03-07 00:00:00 |
| 6          | John Guest         | M           | 1930-06-14 00:00:00 |
| 7          | John Guest II      | M           | 1947-03-09 00:00:00 |
| 8          | John Guest III     | M           | 1965-10-02 00:00:00 |
| 9          | John Guest IV      | M           | 1995-04-25 00:00:00 |
| 10         | Terry Margaret     | F           | 1967-09-07 00:00:00 |
| 11         | Sally Smith        | F           | 1993-05-08 00:00:00 |
| 12         | Patty White        | F           | 1993-06-29 00:00:00 |
| 13         | Aaron Davis        | Unspecified | 1985-08-07 00:00:00 |
| 14         | Jessie Brown       | Unspecified | 1985-05-21 00:00:00 |
| 15         | Quinn Jude         | Unspecified | 1985-04-17 00:00:00 |
| 16         | River Quinn        | Unspecified | 1985-11-11 00:00:00 |
| 25         | James Guest        | M           | 1930-12-01 00:00:00 |
| 26         | James Guest Jr.    | M           | 1930-09-09 00:00:00 |
| 27         | James Guest III    | M           | 1930-10-22 00:00:00 |
| 28         | James Guest IV     | M           | 1995-04-24 00:00:00 |
| 29         | James Margaret     | M           | 1995-09-02 00:00:00 |
| 30         | Jeff Guest         | M           | 1995-06-11 00:00:00 |
| 31         | Jeff Guest II      | M           | 1949-06-17 00:00:00 |
| 32         | Jeff Guest III     | M           | 1949-10-16 00:00:00 |
| 33         | Jeff Guest IV      | M           | 1995-04-30 00:00:00 |
| 34         | Jeff Margaret      | M           | 1949-09-30 00:00:00 |
| 35         | Tom Margaret       | M           | 1949-09-30 00:00:00 |
| 100        | James Vick         | Non-binary  | 2000-03-27 00:00:00 |
| 101        | Travis Albanza     | Non-binary  | 2000-01-31 00:00:00 |
| 102        | Ian Alexander      | Non-binary  | 1994-02-14 00:00:00 |
| 103        | Oilly Alexander    | Non-binary  | 1973-12-12 00:00:00 |
| 226        | Michael Todd       | M           | 1930-06-14 00:00:00 |
| 227        | Janice Mann        | F           | 1947-03-09 00:00:00 |
| 228        | Vanessa Jest       | F           | 1965-10-02 00:00:00 |
| 229        | Johnny Peterson Jr | M           | 1995-04-25 00:00:00 |
| 330        | Mary Margaret      | F           | 1967-09-07 00:00:00 |
| 336        | Richard Guest      | M           | 1930-06-14 00:00:00 |
| 337        | Ricky Guest II     | M           | 1947-03-09 00:00:00 |
| 338        | Bobby Smith        | M           | 1965-10-02 00:00:00 |
| 339        | Bobbert Smith      | M           | 1995-04-25 00:00:00 |
| 340        | Tammy Lammy        | F           | 1967-09-07 00:00:00 |
| 446        | Michella Todd      | F           | 1930-06-14 00:00:00 |
| 447        | Jasmine Robertson  | F           | 1947-03-09 00:00:00 |
| 448        | Andy Jest          | M           | 1965-10-02 00:00:00 |
| 449        | Jonny Peters Jr    | M           | 1995-04-25 00:00:00 |
| 556        | Richardo Ramirez   | M           | 1930-06-14 00:00:00 |

|      |                    |   |                     |
|------|--------------------|---|---------------------|
| 557  | Richy Guest Sr     | M | 1947-03-09 00:00:00 |
| 558  | Waldo Smith        | M | 1965-10-02 00:00:00 |
| 559  | Danny Smiths       | M | 1995-04-25 00:00:00 |
| 666  | Richardson Guest   | M | 1930-06-14 00:00:00 |
| 667  | Ramos Guest II     | M | 1947-03-09 00:00:00 |
| 668  | Bravado Smith      | M | 1965-10-02 00:00:00 |
| 669  | Benjamin Smith     | M | 1995-04-25 00:00:00 |
| 776  | Pamela Todd        | F | 1930-06-14 00:00:00 |
| 777  | Jasmine Johnson    | F | 1947-03-09 00:00:00 |
| 778  | Andrew Jest        | M | 1965-10-02 00:00:00 |
| 779  | Jackson Peters Jr  | M | 1995-04-25 00:00:00 |
| 886  | Richardoz Ramirez  | M | 1930-06-14 00:00:00 |
| 887  | Raoni Guest Sr     | M | 1947-03-09 00:00:00 |
| 888  | Woodrow Smith      | M | 1965-10-02 00:00:00 |
| 889  | Wilson Smiths      | M | 1995-04-25 00:00:00 |
| 4440 | Marlena Margaretta | F | 1967-09-07 00:00:00 |
| 5550 | Samson Ma          | M | 1967-09-07 00:00:00 |
| 6660 | Rita Lammy         | F | 1967-09-07 00:00:00 |
| 7770 | Arlana Margarette  | F | 1967-09-07 00:00:00 |
| 8880 | Samsung Manson     | M | 1967-09-07 00:00:00 |

**FASTPASSTICKET**

| TICKETID | NUMBEROFRIDES | VALIDHOURS |
|----------|---------------|------------|
| 5001     | 10            | 8          |
| 5002     | 8             | 10         |
| 5003     | 5             | 6          |
| 5004     | 7             | 12         |
| 5005     | 3             | 4          |

**FORRIDE**

| RIDENAME      | TICKETID |
|---------------|----------|
| BumperCars    | 5005     |
| BumperCars    | 5115     |
| BumperCars    | 5116     |
| BumperCars    | 6670     |
| BumperCars    | 8890     |
| BumperCars    | 63207    |
| FerrisWheel   | 5002     |
| FerrisWheel   | 5112     |
| FerrisWheel   | 6667     |
| FerrisWheel   | 8887     |
| FerrisWheel   | 63204    |
| HauntedHouse  | 5003     |
| HauntedHouse  | 5113     |
| HauntedHouse  | 6668     |
| HauntedHouse  | 8888     |
| HauntedHouse  | 63205    |
| LogFlume      | 3333     |
| LogFlume      | 5004     |
| LogFlume      | 5114     |
| LogFlume      | 6669     |
| LogFlume      | 8889     |
| LogFlume      | 63206    |
| LogFlume      | 63208    |
| RollerCoaster | 3334     |
| RollerCoaster | 5001     |
| RollerCoaster | 6666     |
| RollerCoaster | 8886     |
| RollerCoaster | 63203    |

**GUEST**

| CUSTOMERID | DATEOFVISIT         |
|------------|---------------------|
| 6          | 2025-05-10 00:00:00 |
| 7          | 2025-06-11 00:00:00 |
| 8          | 2025-07-12 00:00:00 |
| 9          | 2023-07-09 00:00:00 |
| 10         | 2023-07-18 00:00:00 |
| 226        | 2024-08-14 00:00:00 |
| 227        | 2024-08-09 00:00:00 |
| 228        | 2024-08-02 00:00:00 |
| 229        | 2024-08-22 00:00:00 |
| 330        | 2024-08-07 00:00:00 |
| 336        | 2024-08-07 00:00:00 |
| 337        | 2024-08-07 00:00:00 |
| 338        | 2024-09-02 00:00:00 |
| 339        | 2024-09-25 00:00:00 |
| 340        | 2024-10-07 00:00:00 |
| 446        | 2023-10-14 00:00:00 |
| 447        | 2023-10-09 00:00:00 |
| 448        | 2023-10-02 00:00:00 |
| 449        | 2023-10-25 00:00:00 |
| 556        | 2022-11-14 00:00:00 |
| 557        | 2022-11-09 00:00:00 |
| 558        | 2022-11-02 00:00:00 |
| 559        | 2023-11-25 00:00:00 |
| 666        | 2023-12-14 00:00:00 |
| 667        | 2023-12-09 00:00:00 |
| 668        | 2023-12-02 00:00:00 |
| 669        | 2023-12-25 00:00:00 |
| 776        | 2024-01-14 00:00:00 |
| 777        | 2024-01-09 00:00:00 |
| 778        | 2024-01-02 00:00:00 |
| 779        | 2024-01-25 00:00:00 |
| 886        | 2022-02-14 00:00:00 |
| 887        | 2022-02-09 00:00:00 |
| 888        | 2022-02-02 00:00:00 |
| 889        | 2023-03-25 00:00:00 |
| 4440       | 2023-10-07 00:00:00 |
| 5550       | 2024-11-07 00:00:00 |
| 6660       | 2023-01-07 00:00:00 |
| 7770       | 2024-01-07 00:00:00 |
| 8880       | 2019-04-07 00:00:00 |



**HOTEL1**

| POSTALCODE | PROVINCE | CITY      |
|------------|----------|-----------|
| A1B2C3     | BC       | Vancouver |
| B2C3D4     | AB       | Calgary   |
| C3D4E5     | ON       | Toronto   |
| D4E5F6     | QC       | Montreal  |
| E5F6G7     | NS       | Halifax   |

HOTEL2

| HOTELNAME     | MAXGUESTS | POSTALCODE |
|---------------|-----------|------------|
| OceanView     | 200       | A1B2C3     |
| MountainInn   | 150       | B2C3D4     |
| CityCenter    | 300       | C3D4E5     |
| LakesideLodge | 100       | D4E5F6     |
| HarborHotel   | 180       | E5F6G7     |

**HOTELSTAY**

| HOTELBOOKINGID | NUMGUESTS | CHECKINDATE         | CHECKOUTDATE        |
|----------------|-----------|---------------------|---------------------|
| 1              | 2         | 2025-07-10 00:00:00 | 2025-07-12 00:00:00 |
| 2              | 4         | 2025-07-12 00:00:00 | 2025-07-15 00:00:00 |
| 3              | 3         | 2025-07-14 00:00:00 | 2025-07-17 00:00:00 |
| 4              | 2         | 2023-07-16 00:00:00 | 2023-07-18 00:00:00 |
| 5              | 5         | 2023-07-18 00:00:00 | 2023-07-21 00:00:00 |

| HOTELNAME     | ROOMNAME  | BOOKINGNUMBER |
|---------------|-----------|---------------|
| OceanView     | Single    | 1001          |
| MountainInn   | Double    | 1002          |
| CityCenter    | Suite     | 1003          |
| LakesideLodge | Cabin     | 1004          |
| HarborHotel   | Penthouse | 1005          |

**LOYALTYMEMBER**

| CUSTOMERID | LOYALTYID | POINTS | UUID  |
|------------|-----------|--------|-------|
| 1          | 101       | 120    | 10001 |
| 2          | 102       | 80     | 10002 |
| 3          | 103       | 200    | 10003 |
| 4          | 104       | 0      | 10004 |
| 5          | 105       | 30     | 10005 |
| 11         | 106       | 40     | 10006 |
| 12         | 107       | 10     | 10007 |
| 13         | 108       | 64     | 10008 |
| 14         | 109       | 30     | 10009 |
| 15         | 110       | 17     | 10010 |
| 16         | 111       | 2201   | 10011 |
| 25         | 116       | 1      | 20001 |
| 26         | 117       | 3      | 20002 |
| 27         | 118       | 4      | 20003 |
| 28         | 119       | 120    | 20004 |
| 29         | 120       | 111    | 10016 |
| 30         | 121       | 1111   | 10017 |
| 31         | 122       | 443    | 10018 |
| 32         | 123       | 232    | 10019 |
| 33         | 124       | 234    | 10020 |
| 34         | 125       | 355    | 10021 |
| 35         | 126       | 17     | 10022 |
| 100        | 112       | 3341   | 10012 |
| 101        | 113       | 1      | 10013 |
| 102        | 114       | 0      | 10014 |
| 103        | 115       | 995    | 10015 |

**MAINTENANCERECORD1**

| MAINTENANCEPERFORMED | NUMBEROFWORKERS |
|----------------------|-----------------|
| Oil Change           | 2               |
| Brake Inspection     | 3               |
| Structural Check     | 4               |
| Electrical Test      | 2               |
| Safety Audit         | 5               |

MAINTENANCERECORD2

| RIDENAME      | RECORDID | MAINTENANCEPERFORMED | MAINTENANCEDATE     |
|---------------|----------|----------------------|---------------------|
| RollerCoaster | 1        | Oil Change           | 2025-01-10 00:00:00 |
| FerrisWheel   | 1        | Brake Inspection     | 2025-02-20 00:00:00 |
| HauntedHouse  | 1        | Structural Check     | 2025-03-05 00:00:00 |
| LogFlume      | 1        | Electrical Test      | 2025-04-11 00:00:00 |
| BumperCars    | 1        | Safety Audit         | 2025-05-22 00:00:00 |

**OFFERSROOMTYPE**

| HOTELNAME     | ROOMNAME  | QUANTITY |
|---------------|-----------|----------|
| OceanView     | Single    | 20       |
| MountainInn   | Double    | 15       |
| CityCenter    | Suite     | 5        |
| LakesideLodge | Cabin     | 8        |
| HarborHotel   | Penthouse | 2        |



**RATEMODIFIER**

| RATECODE | MODIFIER |
|----------|----------|
| 0        | 1        |
| 1        | 0.9      |
| 2        | 1.1      |
| 3        | 0.8      |
| 4        | 1.2      |

**RIDE**

| RIDENAME      | CAPACITY | INSTALLDATE         |
|---------------|----------|---------------------|
| RollerCoaster | 40       | 2015-06-01 00:00:00 |
| FerrisWheel   | 30       | 2010-04-12 00:00:00 |
| HauntedHouse  | 20       | 2018-10-10 00:00:00 |
| LogFlume      | 25       | 2012-07-21 00:00:00 |
| BumperCars    | 15       | 2011-09-14 00:00:00 |

ROOMTYPE1

| CATEGORY | BASERATE |
|----------|----------|
| Economy  | 100      |
| Standard | 150      |
| Luxury   | 300      |
| Premium  | 500      |
| Rustic   | 120      |

ROOMTYPE2

| ROOMNAME  | MAXGUESTS | CATEGORY |
|-----------|-----------|----------|
| Single    | 1         | Economy  |
| Double    | 2         | Standard |
| Suite     | 4         | Luxury   |
| Penthouse | 6         | Premium  |
| Cabin     | 3         | Rustic   |

**SEASONPASS1**

| SEASONPASSLEVEL | SEASONSTART         | SEASONEND           |
|-----------------|---------------------|---------------------|
| Gold            | 2025-01-01 00:00:00 | 2025-12-31 00:00:00 |
| Silver          | 2025-04-01 00:00:00 | 2025-03-31 00:00:00 |
| Bronze          | 2025-06-01 00:00:00 | 2025-05-31 00:00:00 |
| Platinum        | 2025-02-01 00:00:00 | 2025-01-31 00:00:00 |
| Member          | 2025-03-01 00:00:00 | 2025-02-28 00:00:00 |

**SEASONPASS2**

| UUID | SEASONPASSLEVEL | SEASONSTART         | SEASONEND           | LOYALTYID |
|------|-----------------|---------------------|---------------------|-----------|
| 9001 | Gold            | 2025-01-01 00:00:00 | 2025-12-31 00:00:00 | 101       |
| 9002 | Silver          | 2025-04-01 00:00:00 | 2025-03-31 00:00:00 | 102       |
| 9003 | Bronze          | 2025-06-01 00:00:00 | 2025-05-31 00:00:00 | 103       |
| 9004 | Platinum        | 2025-02-01 00:00:00 | 2025-01-31 00:00:00 | 104       |
| 9005 | Member          | 2025-03-01 00:00:00 | 2025-02-28 00:00:00 | 105       |

**SEASONPASSSPECIAL**

| UUID | RATECODE |
|------|----------|
| 9001 | 1        |
| 9002 | 2        |
| 9003 | 3        |
| 9004 | 4        |
| 9005 | 0        |

**TICKET1**

| VALIDFROM           | VALIDHOURS | VALIDUNTIL          |
|---------------------|------------|---------------------|
| 2025-07-01 00:00:00 | 8          | 2025-07-31 00:00:00 |
| 2025-07-10 00:00:00 | 8          | 2025-07-31 00:00:00 |
| 2025-07-11 00:00:00 | 8          | 2025-07-31 00:00:00 |
| 2025-07-12 00:00:00 | 8          | 2025-07-31 00:00:00 |
| 2025-08-01 00:00:00 | 10         | 2025-08-31 00:00:00 |
| 2025-09-01 00:00:00 | 6          | 2025-09-30 00:00:00 |
| 2025-09-14 00:00:00 | 8          | 2025-09-30 00:00:00 |
| 2025-09-14 00:00:00 | 9          | 2025-09-30 00:00:00 |
| 2025-10-01 00:00:00 | 12         | 2025-10-31 00:00:00 |
| 2025-11-01 00:00:00 | 4          | 2025-11-30 00:00:00 |



**TICKET2**

| TICKETID | BOOKINGNUMBER | VALIDFROM           | REMAININGUSES | VALIDHOURS |
|----------|---------------|---------------------|---------------|------------|
| 3333     | 1002          | 2025-08-01 00:00:00 | 2             | 10         |
| 3334     | 1002          | 2025-08-01 00:00:00 | 1             | 10         |
| 5001     | 1001          | 2025-07-10 00:00:00 | 5             | 8          |
| 5002     | 1002          | 2025-08-01 00:00:00 | 3             | 10         |
| 5003     | 1003          | 2025-09-01 00:00:00 | 2             | 6          |
| 5004     | 1004          | 2025-10-01 00:00:00 | 4             | 12         |
| 5005     | 1005          | 2025-11-01 00:00:00 | 1             | 4          |
| 5112     | 1111          | 2025-07-10 00:00:00 | 5             | 8          |
| 5113     | 1112          | 2025-07-10 00:00:00 | 5             | 8          |
| 5114     | 1113          | 2025-07-11 00:00:00 | 5             | 8          |
| 5115     | 1114          | 2025-07-12 00:00:00 | 5             | 8          |
| 5116     | 1115          | 2025-07-12 00:00:00 | 5             | 8          |
| 6666     | 2223          | 2025-09-14 00:00:00 | 6             | 8          |
| 6667     | 2224          | 2025-09-14 00:00:00 | 7             | 8          |
| 6668     | 2225          | 2025-09-14 00:00:00 | 2             | 8          |
| 6669     | 2226          | 2025-09-14 00:00:00 | 4             | 9          |
| 6670     | 2227          | 2025-09-14 00:00:00 | 4             | 8          |
| 8886     | 2323          | 2025-09-14 00:00:00 | 1             | 8          |
| 8887     | 2424          | 2025-09-14 00:00:00 | 4             | 8          |
| 8888     | 2525          | 2025-09-14 00:00:00 | 3             | 8          |
| 8889     | 2626          | 2025-09-14 00:00:00 | 7             | 8          |
| 8890     | 2727          | 2025-09-14 00:00:00 | 4             | 8          |
| 63203    | 7433          | 2025-09-01 00:00:00 | 2             | 6          |
| 63204    | 7434          | 2025-09-01 00:00:00 | 4             | 6          |
| 63205    | 7435          | 2025-09-01 00:00:00 | 1             | 6          |
| 63206    | 7436          | 2025-09-01 00:00:00 | 2             | 6          |
| 63207    | 7437          | 2025-09-01 00:00:00 | 4             | 6          |
| 63208    | 7438          | 2025-09-01 00:00:00 | 1             | 6          |